

ALLISON LIU

Data Scientist with federal transportation experience translating quantitative analysis and geospatial data into decision-ready tools. Skilled in Python-based GIS workflows, data visualization, and data management.

PROFESSIONAL EXPERIENCE

APRIL 2024 – SEPTEMBER 2025	Data Scientist <i>Bureau of Transportation Statistics, U.S. Department of Transportation</i> Washington, DC - Built a nationally consistent geospatial dataset identifying transportation infrastructure exposed to natural hazards. Developed a custom web map enabling stakeholders to prioritize critical assets for analysis. - Led collaboration with 50+ stakeholders (including state DOTs, local governments, researchers, and industry) to develop a national data standard for bike, pedestrian, and accessibility infrastructure. - Automated geospatial data processing pipelines for federal transportation noise datasets, enabling migration to a lower-cost hosting platform, improving user interface, and increasing the accuracy, reproducibility, and efficiency of statistical updates.
MAY 2022 – MAY 2023	Applied Mathematics Graduate Research Assistant <i>University of Colorado Boulder</i> Boulder, CO - Processed, cleaned, and analyzed seven years of NASA remote sensing imagery. - Applied machine learning, image analysis, and dimensionality reduction techniques to datasets to extract spatial and temporal patterns relevant to solar flare prediction. - Produced reproducible code, maintained datasets, and documented analytical workflows.
FEBRUARY 2021 – DECEMBER 2022	Student Data Scientist <i>Laboratory for Atmospheric and Space Physics (LASP)</i> Boulder, CO - Processed and QA/QC'd 17 years of NASA magnetogram datasets. - Trained and refined generative adversarial networks (GANs), combining historic and current satellite data to create a machine learning ready dataset for solar flare prediction. - Performed feature engineering, evaluated and tuned model architectures, and analyzed statistical performance of model outputs.
JUNE 2017 – AUGUST 2020	Student Research Assistant <i>Kapteyn-Murnane Group, JILA, University of Colorado Boulder</i> Boulder, CO - Designed and built diagnostic instruments for laser beam profile characterization. - Developed MATLAB-based data acquisition interfaces and implemented novel image analysis algorithms.
SEPTEMBER 2020 – AUGUST 2022	Climbing Gym Route Setter at University of Colorado Boulder Boulder, CO - Collaborated with a five-person team to design and set innovative and technically challenging climbing routes for recreational and competition events. - Planned event logistics, managed safety systems, and contributed to a culture of constructive feedback and teamwork in a fast-paced environment.

EDUCATION

2023	M.S. Applied Mathematics, GPA 3.86/4.00 University of Colorado Boulder
2022	B.S. Applied Mathematics, GPA 3.72/4.00 – Cum Laude with Honors University of Colorado Boulder Minor in Computer Science Award: College of Engineering Outstanding Graduate for Research (2022)

SKILLS

GIS & GEOSPATIAL	QGIS, ArcGIS, GDAL, Git, Geospatial Data Processing, Spatial Analysis
PROGRAMMING & ANALYSIS	Python, Git, SQL, Data Visualization, Machine Learning (PyTorch, TensorFlow, scikit-learn), Image Analysis
WEB & TOOLS	HTML/CSS/JavaScript, MS Excel/Word/PowerPoint
LANGUAGES	English – Native proficiency, Chinese (Mandarin) – Limited working proficiency

VOLUNTEERING & OUTREACH

APRIL 2025 – JUNE 2025	Data Analyst @ Bright Block Analyzed bike lane air quality data collected from prototype smart bike lights. Created GIS-ready datasets to support safe cycling advocacy.
MAY 2022 – JUNE 2022	Undergraduate Research Program Mentor <i>Boulder Solar Alliance Research Experience for Undergraduates (REU)</i> Developed and led interactive Python workshops for undergraduate research students.
MAY 2021 – JULY 2021	Machine Learning STEM Camp Instructor Developed curriculum and taught machine learning and data science fundamentals to high school students, fostering interest in computational and applied research.
FEBRUARY 2020 – MAY 2020	STEM Mentor – Elementary Outreach <i>Partnerships for Informal Education in the Community (PISEC)</i> Led hands-on lab activities weekly for primary school students in low-income communities.

PUBLICATIONS & PRESENTATIONS

- Data Augmentation of Magnetograms for Solar Flare Prediction using Generative Adversarial Networks.
A. Liu, W. Carande. *Poster Presented at the American Geophysical Union Conference: New Orleans, LA (2021)*. DOI: 10.1002/essoar.10510080.1
- Generation of extreme-ultraviolet beams with time-varying orbital angular momentum.
L. Rego, K. Dorney, N. Brooks, Q. Nguyen, C. T. Liao, J. San Román, D. Couch, **A. Liu**, E. Pisanty, M. Lewenstein, L. Plaja, H. C. Kapteyn, M. M. Murnane, & C. Hernández-García. *Science* 364, 6447 (2019). DOI: 10.1126/science.aaw9486
- Detection of the Keto-Enol Tautomerization in Acetaldehyde, Acetone, Cyclohexanone, and Methyl Vinyl Ketone with a Novel VUV Light Source.
D. Couch, Q. Nguyen, **A. Liu**, D. Hickstein, H. Kapteyn, M. Murnane, & N. Labbe. *Proc. Combust. Inst.* 38 (2021). DOI: 10.1010/j.proci.2020.06.139